CAPSTONE PROJECT · DATA ANALYTICS

Risk Analytics & Claims Intelligence

AegisLife INSURANCE PVT. LTD.

End-to-End Insurance Analytics using **SQL, Python, Excel & Power BI**

MYSQL | PYTHON | EXCEL | POWER BI

Prepared by **Anuj Magre**

BUSINESS CONTEXT

Business Challenges at AegisLife

Claims Monitoring
Track premium and claim performance across products and regions

Fraud Detection
Identify suspicious patterns and high-risk claims

Performance Analytics
Measure agent productivity and regional profitability

Risk Analysis
Segment customers by risk profile and exposure

PROJECT SCOPE

Project Objectives

Data Quality Validation

Validate integrity, completeness, and consistency across all source datasets.

Claims & Fraud Analysis

Uncover patterns in claim behavior, fraud signals, and risk segmentation.

Business KPI Measurement

Compute premium, loss ratio, claim frequency, and agent performance metrics.

Statistical Validation

Apply hypothesis testing to validate business assumptions with statistical rigor.

Claims Forecasting

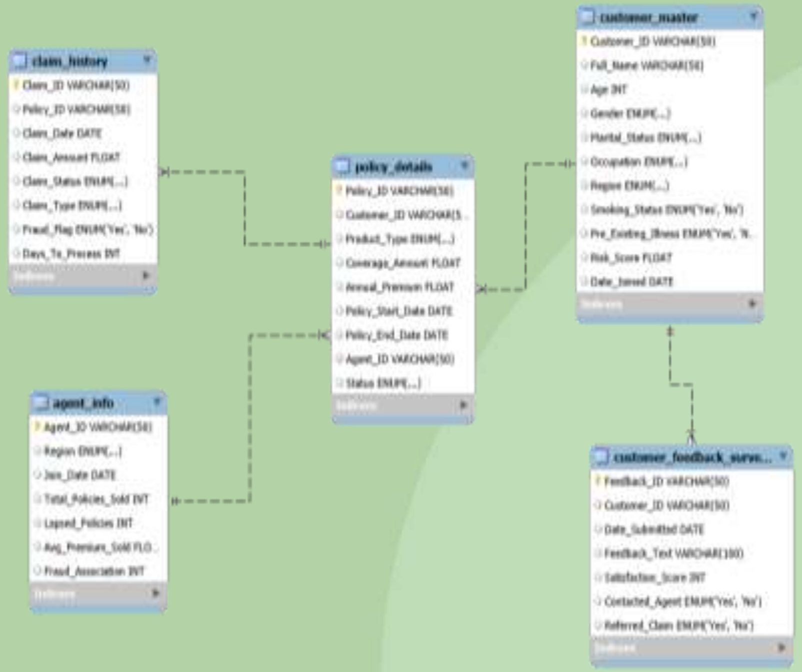
Predict future claim volumes to support reserve planning and risk budgeting.

Interactive Dashboards

Build Power BI reports enabling self-service analytics for stakeholders.

DATA FOUNDATION

Data Model & Dataset Overview



Five Core Datasets

- **Customer Master** — Customer demographics and segmentation
- **Policy Details** — Coverage, premium & tenure
- **Claim History** — Status, amount & type
- **Agent Info** — Performance & regional data
- **Customer Feedback** — Satisfaction scores

1,648
Customers

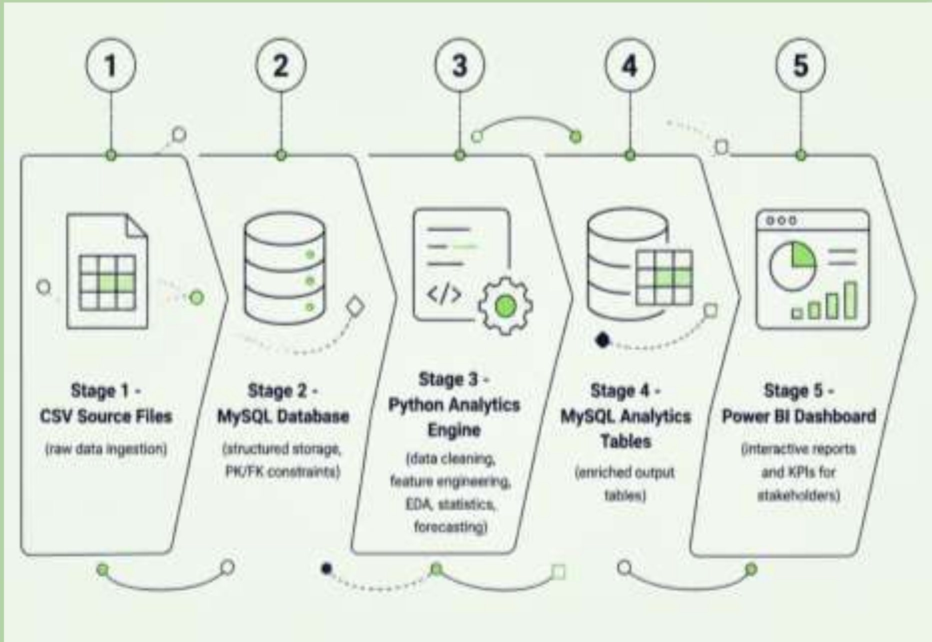
2,828
Policies

1,407
Claims

300
Agents

ARCHITECTURE

End-to-End Analytics Pipeline



Pipeline Design Principles

- Raw CSV ingestion into normalized MySQL schema
- Python handles all transformation, EDA, and modeling
- Clean analytics tables written back to MySQL
- Power BI connects live for interactive reporting

Python Modules: Data Cleaning · Feature Engineering · EDA · Statistics · Forecasting

DATA ENGINEERING

Data Validation & Feature Engineering

Data Quality & Validation

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PK / FK Validation

Validated all primary and foreign key relationships to ensure accurate joins and analytical consistency across the insurance data model.

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Datatype Conversion

Standardized date, numeric, and categorical fields for consistent downstream analysis.

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Referential Integrity Checks

Verified all policies, claims, customers, and agents were correctly linked with zero orphan records.

Business Features Engineered

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- **Policy Duration** — Tenure in days from issue date
- **Premium Collected** — Cumulative premium per policy
- **Customer Risk Segment** — Low / Moderate / High classification
- **Customer Age Group** — 18–30, 31–45, 46–60, 60+
- **Agent Fraud Ratio** — Fraudulent claims per agent
- **Claim Severity** — Claim amount vs. sum assured ratio
- **Underwriting Loss** — Premium vs. payout differential
- **Future Risk Score** — Composite risk indicator for forecasting

ANALYTICS

Exploratory Data Analysis — Key Findings

What the Data Revealed

- **Health Insurance** generated the highest total claim costs
- **South Region** recorded the highest loss ratio, signaling underwriting risk
- **Moderate-risk customers** dominate the portfolio — largest segment by volume
- **Fraud patterns** vary significantly by region — North shows elevated fraud rate

These insights directly informed risk segmentation strategy and regional underwriting recommendations.

STATISTICAL RIGOR

Hypothesis Testing & Statistical Validation

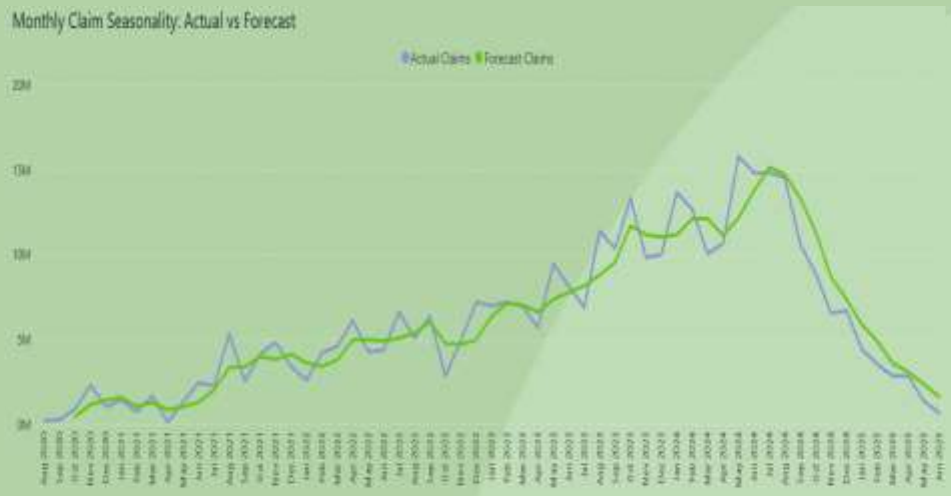
Business decisions were validated using statistical tests — not visual observations alone.

Decision Rule: Reject H_0 if p-value < 0.05

Statistical Test	Business Question	Outcome
T-Test	Do Smokers file significantly higher claims than Non-Smokers?	✓ Significant
ANOVA	Do claim amounts differ significantly across regions?	✓ Significant
Chi-Square	Is Claim Status independent of Claim Type?	✓ Significant
Correlation	Is Risk Score linearly related to Claim Amount?	✓ Positive

FORECASTING

Claim Forecasting & Reserve Planning



3-Month Moving Average Model

- Smooths short-term volatility to reveal underlying claim trends
- Forecast indicates claims approaching reserve threshold by August
- Supports proactive capital allocation and reinsurance decisions

Trend

Upward claim trajectory confirmed

Action

Increase reserves for Q3–Q4

POWER BI DASHBOARD SUITE

Executive Dashboard Suite

Six interactive pages delivering real-time visibility across claims, fraud, customer risk, and regional performance.

Executive Overview

Portfolio-level KPIs and financial performance

Claims & Fraud Analysis

Fraud flag rates and claims cost breakdowns

Customer Risk Analysis

Risk tiering and customer segmentation

Regional Analysis

Loss ratios and performance by region

Agent Performance

Productivity and claim quality by agent

Forecasting Dashboard

Predictive modeling for reserve planning

KEY FINDINGS

Top 5 Business Insights

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130.8%

Loss Ratio

Claims paid ₹127.39M vs. premiums collected ₹97.39M

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₹127M

Total Claims Paid

Highest claim costs across all insurance products

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159.3%

South Region Loss

Highest regional loss ratio — critical underwriting concern

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53.3%

North Fraud Ratio

Highest fraud concentration among all regions

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49%

Fraud-Flagged Claims

Nearly half of all claims flagged for fraud review

STRATEGIC RECOMMENDATIONS

Strategic Recommendations

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Review Health Insurance Pricing

Reassess premiums to restore profitability

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Strengthen Fraud Detection

Deploy ML-based models to reduce fraud losses

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Reassess South Region Underwriting

Tighten risk criteria in the highest-loss region

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Monitor High-Risk Customers

Proactively track high-risk profiles

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Use Forecasting for Reserve Planning

Apply predictive models for reserve accuracy

Thank You

This project demonstrates an end-to-end analytics workflow — covering SQL, Python, Statistics, Forecasting, and Power BI — enabling data-driven decision making at AegisLife Insurance.

SQL

Data extraction & transformation

Python

Analysis & modeling

Statistics

Hypothesis testing & inference

Forecasting

Predictive trend modeling

Power BI

Interactive dashboards

Prepared By: Anuj Magre

Questions & Discussion — Open for feedback, deep dives, and methodology questions.